

Same Estimate, Different Information

Belief Compression in Optimal Execution under Latent Liquidity

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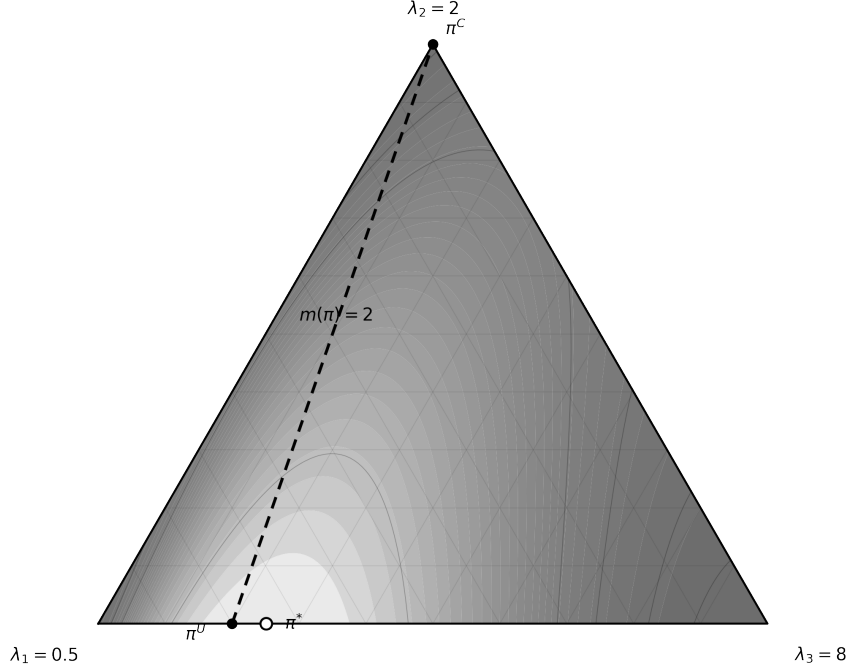


Figure 1: Belief simplex for the three-state baseline $\lambda = (0.5, 2, 8)^\top$. The grayscale field encodes the exact certainty-equivalent decision loss $R_{CE}(\pi, 1)$ at the first learning-sensitive horizon. Five contour levels, one from each stable topological regime of the baseline loss surface, produce nine visible connected components with counts 1, 2, 3, 2, 1. The dashed segment is the fixed-mean fiber $m(\pi) = 2$. Its endpoints $\pi^C = (0, 1, 0)$ and $\pi^U = (0.8, 0, 0.2)$ share the same current expected impact and, under the affine condition, the same unconditional future mean-impact path while differing in the value of future adaptation. The open marker π^* denotes the baseline global worst belief on the low/high support pair.

Abstract

This paper studies an execution problem in which temporary price impact is driven by a finite-state latent liquidity regime. The trader observes a Bayesian posterior over regimes, while a downstream execution rule may receive only the posterior expected impact. The question is whether two posterior states that report the same liquidity estimate should generate the same trade.

The analysis separates two channels through which mean compression can fail. First, the mean may fail to determine expected future liquidity. Second, even when the entire unconditional expected-liquidity path is matched, posterior shape can change the value of adapting future trades to new information. The second case is isolated by imposing affine mean dynamics, $P\lambda = \alpha\mathbf{1} + \beta\lambda$. At the first learning-sensitive horizon, the effect of future information is summarized by a Jensen-gap premium $J(\pi)$. The corresponding certainty-equivalent action distortion and exact decision loss are derived for the two-decision horizon when the random next posterior mean is replaced by its conditional expectation.

For a symmetric finite-state signal experiment, $J(\pi)$ is convex along every set of beliefs with a common current mean. Since the exact certainty-equivalent loss is increasing and convex in J ,

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a worst-case belief for any reported liquidity estimate can be chosen at an extreme point of that mean-constrained belief set, hence with support on at most two liquidity regimes. The resulting worst-case loss is therefore obtained by a finite comparison of pairwise regime mixtures. A three-state example shows that beliefs with identical current and unconditional future mean-impact paths can still produce different optimal trades, and it also shows that the minimum/maximum-regime pair need not be pointwise worst for every reported mean.

Keywords: optimal execution; partial information; hidden Markov models; latent liquidity; Bayesian filtering; belief compression; certainty equivalence; price impact.

JEL Classification: C61, D83, G11, G12.

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1 Introduction

Partial information is natural in optimal execution: liquidity, order-flow conditions, latent alpha, and price-impact parameters are not observed without noise. A standard solution is to filter the hidden state and optimize using the posterior belief as part of the state variable. The present paper asks a different question. Suppose an execution layer receives only a scalar estimate of latent temporary impact,

$$m(\pi) = \pi\lambda.$$

When is this estimate enough for the execution decision, and when it is not, how large can the resulting decision loss be? Figure 1 summarizes the geometric object studied below: the posterior simplex, the exact certainty-equivalent loss field on that simplex, and the fixed-mean fiber on which equal reported liquidity coexists with different posterior shapes.

The distinction matters because a point estimate can preserve a forecast while discarding the information structure behind it. Two beliefs can have the same posterior expected impact but different shapes. The obvious reason they might require different trades is that they imply different expected future liquidity. That explanation is removed here. Under an affine restriction on the hidden-state transition law, equal current means imply equal unconditional expected impact at every future horizon. Any remaining difference in optimal execution is then attributable to the value of adapting to future signals, rather than to a difference in the expected liquidity path.

The main quantitative object is a certainty-equivalent (CE) decision rule. The two-decision horizon $h = 2$ is analytically central rather than an arbitrary truncation: under the affine mean condition, $h = 1$ is mean-sufficient, so $h = 2$ is the first horizon at which a future signal can change the current execution decision. At that first learning-sensitive horizon, the CE rule replaces the random next posterior mean by its conditional expectation. The paper derives the action distortion and the exact current-decision loss generated by that replacement. It then asks an adversarial question: conditional on a reported liquidity estimate m , which posterior belief makes the CE approximation most costly?

The answer is finite. Under a symmetric observation experiment, the relevant information premium is convex on a fixed-mean posterior fiber. The exact CE loss is an increasing convex function of that premium. Standard finite-dimensional extreme-point geometry then implies that a worst-case posterior can be chosen with support on at most two hidden states. Thus a continuum optimization over the posterior simplex reduces to a comparison of finitely many pairwise mixtures.

The aim is not to develop a new general theory of POMDP compression. Value-directed belief compression and invariant-subspace conditions are established in the POMDP literature [11]. Convexity and convex-order effects of Bayesian updating are also general phenomena [3], and CE policies can be suboptimal in general partially observed systems [4]. The contribution here is the execution-specific construction: matched expected-liquidity paths, an exact CE decision-loss formula, and an exact fixed-estimate worst-case characterization at the first learning-sensitive horizon under a tractable symmetric-signal model.

Main results. The paper establishes the following sequence.

1. At the penultimate decision, posterior expected impact is sufficient for the execution action exactly when $P\lambda$ is affine in λ . This is best viewed as an execution specialization of known linear belief-compression structure.

2. Under that affine condition, equal current means generate the same unconditional expected impact at every future horizon.
3. Informative observations can nevertheless make equal-mean beliefs imply different optimal trades. The difference is represented by a Jensen-gap information premium $J(\pi)$.
4. The CE action over-executes relative to the belief-aware optimum whenever $J(\pi) > 0$, and its exact current-decision value loss at $h = 2$ is

$$R_{\text{CE}}(\pi, q) = q^2 \frac{J(\pi)^2 m(\pi)^2}{[m(\pi) + B(m(\pi))]^2 [m(\pi) + B(m(\pi)) - J(\pi)]}.$$

5. For a symmetric finite-state signal experiment, J is convex on each fixed-mean belief fiber. Consequently, a maximum CE loss at a reported mean can be attained at a posterior supported on at most two regimes and is found by a finite pairwise comparison.

2 Related Work

Partial-information and regime-switching execution models provide the closest finance context. Bayraktar and Ludkovski [2] consider Markov-modulated order flow in an execution problem. Colaneri et al. [8] formulate liquidation with price impact and an unobservable finite-state Markov chain, reducing the problem by filtering. Casgrain and Jaimungal [5] solve trading problems with learning in latent-alpha models. Chevalier, Hafsi, and Ly Vath [6] study incomplete-information execution with hidden liquidity in a Hawkes-based order-flow model. Chevalier et al. [7] study execution under stochastic regime-switching liquidity and general transient price impact, without the posterior-compression question considered here. Dammann and Ferrari [9] analyze liquidation with an unobservable Bernoulli return parameter. In a different liquidation problem with regime-switching volatility, Vaicenavicius [12] shows that, for priors with more than two support points, time, posterior mean, and instantaneous volatility are no longer sufficient without an additional learning-time statistic.

Learning and regret in execution are also prior topics. Park and Van Roy [10] study adaptive execution while learning price impact and derive a finite-time expected regret bound for an adaptive CE policy. Bank, Cartea, and Körber [1] quantify the monetary value of trade signals and show that additional information can change execution timing and volume. These works rule out broad claims that learning, the value of information, certainty equivalence, or regret are new to execution.

The relevant general control literature is equally important. Poupart and Boutilier [11] connect lossless linear POMDP compression to reward-containing invariant subspaces. Bozkurt et al. [4] provide general suboptimality bounds for CE policies in partially observed systems. Bayraktar and Wang [3] prove convex-order and convexity-preservation results for Bayesian posterior statistics. Accordingly, neither the affine compression condition nor the generic Jensen/convex-order mechanism below is claimed as new general mathematics.

The paper instead isolates a specific execution question not directly characterized in these works: among latent-liquidity posteriors that report the same expected impact, what is the exact loss from using the same CE execution action, and how can the worst compatible posterior be characterized?

3 Model

Let

$$S_t \in \{1, \dots, n\}$$

be a hidden Markov chain with row-stochastic transition matrix P . State i has temporary-impact coefficient $\lambda_i > 0$, collected in the column vector λ . The trader's posterior belief is the row vector

$$\pi_t(i) = \mathbb{P}(S_t = i \mid \mathcal{F}_t), \quad m(\pi_t) = \pi_t \lambda.$$

At a nonterminal date, with remaining inventory $q \geq 0$, the trader executes the fraction $x \in [0, 1]$. Conditional on the current hidden state, expected temporary-impact cost is

$$\lambda_{S_t} (xq)^2.$$

Remaining inventory incurs the quadratic cost

$$\rho((1-x)q)^2, \quad \rho > 0.$$

After the decision, the hidden state transitions according to P . A passive signal Y is then drawn conditionally on the new state, with likelihood $O_j(y) = \mathbb{P}(Y = y \mid S_{t+1} = j)$. The execution action does not affect the transition or observation law. Bayes' rule gives the next posterior π' . All inventory remaining at the terminal date is forcibly liquidated.

The impact term should be read as a conditional expected temporary-impact cost. The model does not assume that observing the realized execution cost mechanically reveals S_t .

Let h denote the number of nonterminal decisions remaining before terminal liquidation. The terminal normalized value is

$$v_0(\pi) = m(\pi).$$

Quadratic homogeneity gives

$$V_h(q, \pi) = q^2 v_h(\pi).$$

Define

$$K_h(\pi) = \rho + \mathbb{E}[v_{h-1}(\pi') \mid \pi].$$

The Bellman recursion is

$$v_h(\pi) = \min_{0 \leq x \leq 1} \{m(\pi)x^2 + K_h(\pi)(1-x)^2\}.$$

Since $m(\pi) > 0$ and $K_h(\pi) > 0$, the optimizer is interior:

$$x_h^*(\pi) = \frac{K_h(\pi)}{m(\pi) + K_h(\pi)}, \quad v_h(\pi) = \frac{m(\pi)K_h(\pi)}{m(\pi) + K_h(\pi)}. \quad (1)$$

4 A Mean-Sufficiency Benchmark

Call the posterior mean decision-sufficient at horizon h if there is a function φ_h such that

$$x_h^*(\pi) = \varphi_h(m(\pi))$$

for every posterior belief.

Proposition 1 (Penultimate decision). *Assume λ is nonconstant. At $h = 1$, the posterior mean is decision-sufficient for every belief if and only if*

$$P\lambda = \alpha \mathbf{1} + \beta \lambda \quad (2)$$

for some constants α, β .

Proof. At $h = 1$,

$$K_1(\pi) = \rho + \pi P\lambda$$

and therefore

$$x_1^*(\pi) = \frac{\rho + \pi P\lambda}{m(\pi) + \rho + \pi P\lambda}.$$

For fixed $m(\pi)$ the right-hand side is strictly increasing in $\pi P\lambda$. Thus mean sufficiency is equivalent to the linear functional $\pi \mapsto \pi P\lambda$ being constant on every intersection of the simplex with a level set of $\pi \mapsto \pi \lambda$. To see necessity, take an interior belief and any direction h satisfying $h^\top \mathbf{1} = h^\top \lambda = 0$. Small perturbations in both directions remain inside the same mean fiber, so $h^\top P\lambda = 0$. Hence $P\lambda$ is orthogonal to the null space generated by the normalization and mean constraints, which is equivalent to $P\lambda \in \text{span}\{\mathbf{1}, \lambda\}$. Sufficiency is immediate. $\square$

Proposition 1 is closely aligned with general invariant- subspace conditions for lossless linear belief compression [11]; its role here is to create a controlled execution environment in which the entire expected-liquidity path can be matched.

Corollary 1 (Mean-path equivalence). *If (2) holds, then for every $k \geq 1$,*

$$P^k \lambda = \alpha \left(\sum_{j=0}^{k-1} \beta^j \right) \mathbf{1} + \beta^k \lambda.$$

Hence

$$m(\pi^A) = m(\pi^B) \implies \pi^A P^k \lambda = \pi^B P^k \lambda \quad \text{for all } k \geq 1.$$

Proof. Use $P\mathbf{1} = \mathbf{1}$ and induction. $\square$

Thus any execution difference between equal-mean beliefs under (2) cannot be attributed to a different unconditional expected impact path.

Proposition 2 (No-learning benchmark). *Suppose future observations are uninformative about the hidden state. Then the posterior mean is decision-sufficient at every horizon if and only if (2) holds.*

Proof. Necessity follows from Proposition 1. Under uninformative observations, $\pi' = \pi P$ deterministically. If (2) holds,

$$m(\pi P) = \alpha + \beta m(\pi).$$

Backward induction from $v_0(\pi) = m(\pi)$ in (1) then shows that v_h and x_h^* are functions of the posterior mean alone at every horizon. $\square$

5 Learning-Induced Failure of Mean Compression

Assume (2) and set

$$A = \rho + \alpha.$$

At $h = 1$ the normalized value is

$$v_1(\pi) = g(m(\pi)), \quad g(m) = \frac{m(A + \beta m)}{A + (1 + \beta)m}. \quad (3)$$

The denominator is positive on the reachable mean interval because it equals $m + K_1$. Direct differentiation gives

$$g''(m) = -\frac{2A^2}{[A + (1 + \beta)m]^3} \leq 0. \quad (4)$$

The inequality is strict when $A \neq 0$.

At $h = 2$, let

$$M' = m(\pi')$$

be the random posterior mean after the transition and signal. Its conditional expectation is

$$\mathbb{E}[M' \mid \pi] = \pi P \lambda = \alpha + \beta m(\pi).$$

Define the information premium

$$J(\pi) = g(\mathbb{E}[M' \mid \pi]) - \mathbb{E}[g(M') \mid \pi]. \quad (5)$$

By (4),

$$J(\pi) \geq 0.$$

The quantity $J(\pi)$ is the reduction in expected normalized continuation cost obtained by conditioning the next execution decision on the realized posterior mean rather than replacing that random mean by its expectation. If $A \neq 0$, the inequality is strict whenever M' is nondegenerate.

Set

$$B(m) = \rho + g(\alpha + \beta m).$$

Then

$$K_2(\pi) = B(m(\pi)) - J(\pi) \quad (6)$$

and

$$x_2^*(\pi) = \frac{B(m(\pi)) - J(\pi)}{m(\pi) + B(m(\pi)) - J(\pi)}. \quad (7)$$

Proposition 3 (Adaptation value). *For fixed current mean m ,*

$$\frac{\partial x_2^*}{\partial J} = -\frac{m}{[m + B(m) - J]^2} < 0.$$

Thus a larger value of future information causes the belief-aware trader to execute less inventory immediately.

The sign has a simple interpretation. Future signals allow the trader to condition the next trade on the realized posterior. More valuable future adaptation therefore creates an option value of retaining inventory.

6 Exact Certainty-Equivalent Decision Loss at $h = 2$

Define the current CE action by replacing the random next posterior mean in the continuation calculation by its conditional expectation:

$$x_2^{\text{CE}}(m) = \frac{B(m)}{m + B(m)}. \quad (8)$$

This is a decision-interface compression. The current execution rule uses only the scalar mean. At the next date an upstream filter may supply the newly computed posterior mean; since $h = 1$ is mean-sufficient under (2), continuation from that point is exact. The loss below is therefore an exact current-decision CE loss at the two-decision horizon. It is not a claim that a filter which permanently discards the full posterior can reconstruct future posterior means from the previous mean alone.

Proposition 4 (Action distortion). *For any belief at $h = 2$,*

$$\Delta_x(\pi) := x_2^{\text{CE}}(m(\pi)) - x_2^*(\pi) = \frac{m(\pi)J(\pi)}{[m(\pi) + B(m(\pi))][m(\pi) + B(m(\pi)) - J(\pi)]}. \quad (9)$$

Hence $J(\pi) > 0$ implies $x_2^{\text{CE}} > x_2^$: the CE action over-executes relative to the belief-aware optimum.*

Theorem 1 (Exact CE decision loss at $h = 2$). *For inventory q , the value loss from using x_2^{CE} at the current date and then continuing optimally is*

$$R_{\text{CE}}(\pi, q) = q^2 \frac{J(\pi)^2 m(\pi)^2}{[m(\pi) + B(m(\pi))]^2 [m(\pi) + B(m(\pi)) - J(\pi)]}. \quad (10)$$

Proof. For fixed π , the normalized current objective is

$$Q_\pi(x) = mx^2 + [B(m) - J](1 - x)^2.$$

Completing the square gives

$$Q_\pi(x) - Q_\pi(x_2^*) = [m + B(m) - J](x - x_2^*)^2.$$

Substituting (9) and restoring the factor q^2 yields (10). $\square$

For small J ,

$$\Delta_x = O(J), \quad R_{\text{CE}} = O(J^2).$$

Thus noticeable policy distortion can coexist with a comparatively small value loss, a direct consequence of second-order optimality around x_2^* .

7 Worst-Case Loss at a Fixed Reported Mean

The observation experiment is now specialized. Assume $n \geq 2$, let the signal alphabet equal the hidden-state alphabet, and take $\eta \in [0, 1]$ with

$$O_j(y) = \begin{cases} \eta, & y = j, \\ c, & y \neq j, \end{cases} \quad c = \frac{1 - \eta}{n - 1}, \quad d = \eta - c. \quad (11)$$

The uninformative experiment is $\eta = 1/n$, for which $d = 0$. The case $\eta = 1$ gives perfect classification. Values $\eta < 1/n$ are negatively diagnostic rather than uninformative.

Relabel the hidden states, without loss of generality, so that $\lambda_1 < \dots < \lambda_n$. Fix $m \in [\lambda_1, \lambda_n]$ and define

$$\mathcal{F}_m = \{\pi \in \Delta^{n-1} : \pi\lambda = m\}.$$

Under (2), the predictive belief

$$r = \pi P$$

has the common mean

$$\mu = r\lambda = \alpha + \beta m$$

for every $\pi \in \mathcal{F}_m$.

For signal y ,

$$p_y = \mathbb{P}(Y = y \mid \pi) = c + dr_y \quad (12)$$

and, whenever $p_y > 0$,

$$M_y = \mathbb{E}[\lambda_{S_{t+1}} \mid Y = y, \pi] = \frac{c\mu + dr_y\lambda_y}{c + dr_y}. \quad (13)$$

Assume first that

$$s := 1 + \beta \neq 0.$$

Equation (3) admits the decomposition

$$g(x) = \frac{\beta}{s}x + \frac{A}{s^2} - \frac{A^2}{s^2} \frac{1}{A + sx}. \quad (14)$$

Let

$$D_\mu = A + s\mu, \quad D_y = A + s\lambda_y.$$

Since the affine and constant terms in (14) cancel in the Jensen gap,

$$J(\pi) = \frac{A^2}{s^2} \left[\sum_y \frac{(c + dr_y)^2}{cD_\mu + dr_yD_y} - \frac{1}{D_\mu} \right]. \quad (15)$$

Lemma 1 (Fixed-mean convexity). *For the symmetric signal experiment (11), $J(\pi)$ is convex on $\mathcal{F}_m$.*

Proof. For fixed m , D_μ is constant. Define

$$\psi_y(z) = \frac{(c + dz)^2}{cD_\mu + dzD_y}.$$

On points with positive signal probability,

$$\psi_y''(z) = \frac{2c^2d^2(D_\mu - D_y)^2}{(cD_\mu + dzD_y)^3} \geq 0. \quad (16)$$

The denominator is positive because it equals $p_y[A + sM_y]$ and the Bellman denominator $A + sM_y$ is positive on the reachable mean interval. Zero-probability boundary terms are defined by continuous extension. Hence (15) is convex in the predictive belief r . Since $r = \pi P$ is affine in π , the result follows. $\square$

At $\eta = 1/n$, $J \equiv 0$. At $\eta = 1$, the expression is affine in r on a fixed-mean fiber, so convexity remains valid. If $\beta = -1$, the reciprocal decomposition (14) is singular, but the result follows directly. In that case positivity of the Bellman denominator implies $A > 0$ and

$$g(x) = \frac{x(A - x)}{A}, \quad J(\pi) = \frac{1}{A} \left[\sum_y \frac{(c\mu + dr_y\lambda_y)^2}{c + dr_y} - \mu^2 \right].$$

For each summand, with $z = r_y$ and positive signal probability,

$$\frac{d^2}{dz^2} \frac{(c\mu + dz\lambda_y)^2}{c + dz} = \frac{2c^2d^2(\lambda_y - \mu)^2}{(c + dz)^3} \geq 0,$$

with zero-probability boundary cases again obtained by continuity. Thus the fixed-mean convexity conclusion also holds for $\beta = -1$.

Remark 1 (Role of the symmetric signal experiment). The symmetry in (11) is substantive for the convexity argument. It yields the one-coordinate forms $p_y = c + dr_y$ and (13), which reduce (15) to a sum of convex functions of the individual predictive probabilities r_y . For a general state-dependent likelihood matrix, each signal probability and posterior mean can couple several coordinates of r , so the proof above does not establish fixed-mean convexity. No asymmetric-signal extension is claimed here; identifying conditions under which an analogous reduction remains valid is left open.

For fixed m , write

$$C = m + B(m).$$

The action distortion depends on J through

$$\Delta_x = \frac{m}{C} \frac{J}{C - J},$$

which is increasing and convex for $0 \leq J < C$. Likewise, apart from the positive constant $q^2 m^2 / C^2$, the value loss depends on

$$\phi(J) = \frac{J^2}{C - J},$$

for which

$$\phi'(J) = \frac{J(2C - J)}{(C - J)^2} > 0, \quad \phi''(J) = \frac{2C^2}{(C - J)^3} > 0. \quad (17)$$

Theorem 2 (Fixed-mean worst-case CE loss). *Assume $\lambda_1 < \dots < \lambda_n$. Under (2) and the symmetric signal experiment, at least one maximizer of J , Δ_x , and R_{CE} over $\mathcal{F}_m$ is supported on at most two hidden states.*

For every pair $i < j$ satisfying $\lambda_i \leq m \leq \lambda_j$, define

$$\pi_i^{ij}(m) = \frac{\lambda_j - m}{\lambda_j - \lambda_i}, \quad \pi_j^{ij}(m) = \frac{m - \lambda_i}{\lambda_j - \lambda_i}, \quad (18)$$

with all other coordinates zero. Then

$$J_{\max}(m) = \max_{\substack{i < j \\ \lambda_i \leq m \leq \lambda_j}} J(\pi^{ij}(m)), \quad (19)$$

and

$$R_{\max}(m, q) = q^2 \frac{J_{\max}(m)^2 m^2}{[m + B(m)]^2 [m + B(m) - J_{\max}(m)]}. \quad (20)$$

Proof. Lemma 1 and (17) imply that J , Δ_x , and R_{CE} are continuous convex functions on the compact polytope $\mathcal{F}_m$. A continuous convex function on a polytope has at least one extreme-point maximizer.

It remains to characterize the extreme points. Suppose a feasible posterior has at least three positive coordinates. On those coordinates there exists a nonzero vector h satisfying

$$\mathbf{1}^\top h = 0, \quad \lambda^\top h = 0.$$

For sufficiently small $\varepsilon > 0$, both $\pi + \varepsilon h$ and $\pi - \varepsilon h$ are feasible, so π is not extreme. Hence every extreme point has support on at most two states. Solving normalization and mean constraints on a two-state support gives (18). Because (10) is increasing in J for fixed m , the same pair maximizes J and CE loss. $\square$

The support argument is the finite-state specialization of classical extreme-point results for moment-constrained probability sets; see, for example, Winkler [13]. The elementary proof is included above to keep the reduction self-contained.

The theorem does not say that every maximizer is extreme, nor does it say that the winning pair must be the lowest- and highest-impact regimes.

8 Three-State Baseline

Consider

$$\lambda = (0.5, 2, 8)^\top$$

and

$$P = \begin{pmatrix} 0.70 & 0.30 & 0 \\ 0.20 & 0.75 & 0.05 \\ 0 & 0.30 & 0.70 \end{pmatrix}.$$

Then

$$P\lambda = 0.61 + 0.7\lambda.$$

With $\rho = 1$, the fixed point of the mean recursion $m \mapsto 0.6 + 0.7m$ is $m = 2$.

Take

$$\pi^C = (0, 1, 0), \quad \pi^U = (0.8, 0, 0.2).$$

Both have current mean 2, and Corollary 1 implies that they have the same unconditional expected impact at every future horizon. Because 2 is the fixed point, that common path is identically 2.

Use the symmetric observation experiment with $\eta = 0.8$. Figure 1 displays the corresponding loss field and the fixed-mean fiber $m(\pi) = 2$. The two next-posterior-mean distributions are different despite their common expectation. The information premia are

$$J(\pi^C) \approx 0.00973268, \quad J(\pi^U) \approx 0.04203743.$$

The CE action at $m = 2$ is

$$x_2^{\text{CE}}(2) = \frac{11}{21} \approx 0.52380952.$$

The belief-aware actions are

$$x_2^*(\pi^C) \approx 0.52270348, \quad x_2^*(\pi^U) \approx 0.51899519.$$

Thus π^U , which has the larger information premium, preserves more inventory for future adaptation.

For π^U and unit inventory,

$$R_{\text{CE}}(\pi^U, 1) \approx 9.63725 \times 10^{-5}.$$

By Theorem 2, the entire $m = 2$ fiber need only be checked at its two endpoints, π^C and π^U . Since $J(\pi^U) > J(\pi^C)$, π^U is the global worst posterior compatible with the reported mean $m = 2$.

Exact belief-tree recursion also shows that the action difference is not confined to the two-step calculation.

Table 1: Initial optimal fraction under the two equal-mean baseline beliefs.

Nonterminal decisions h	$x_h^*(\pi^C)$	$x_h^*(\pi^U)$	Gap
1	0.600000	0.600000	0.000000
2	0.522703	0.518995	0.003708
5	0.496614	0.487489	0.009125
8	0.495888	0.486300	0.009588

On the grid

$$\eta \in \{1/3, 0.5, 0.7, 0.8, 0.9, 1\}, \quad \rho \in \{0.25, 0.5, 1, 2, 4\}, \quad h \in \{1, 2, 3, 5, 8\},$$

the gap is zero up to numerical precision at the uninformative signal $\eta = 1/3$ and is positive in all 100 informative cases with $h \geq 2$. These computations are robustness checks, not monotonicity theorems.

8.1 Global Worst Case in the Baseline Model

Theorem 2 converts the global search into three one-dimensional pairwise problems. Numerical maximization of those three curves for $\rho = 1$ and $\eta = 0.8$ gives the largest normalized CE loss on the low/high support pair at approximately

$$m^* = 2.38046267, \quad \pi^* \approx (0.74927162, 0, 0.25072838),$$

with

$$J(\pi^*) \approx 0.04293911$$

and

$$R_{\text{CE}}(\pi^*, 1) \approx 1.01246 \times 10^{-4}.$$

8.2 Failure of Universal Low/High Dominance

It is tempting to infer that the lowest- and highest-impact states must always form the worst pair. That inference is false.

Keep the same λ , P , $\rho = 1$, and $\eta = 0.8$, but fix the reported mean at $m = 7.4$. Two feasible extreme posteriors are

$$\pi^{13} = (0.08, 0, 0.92)$$

and

$$\pi^{23} = (0, 0.10, 0.90).$$

Their information premia are

$$J(\pi^{13}) \approx 0.00876996, \quad J(\pi^{23}) \approx 0.00946150.$$

Their normalized CE losses are approximately

$$R_{\text{CE}}(\pi^{13}, 1) \approx 2.9555 \times 10^{-6},$$

$$R_{\text{CE}}(\pi^{23}, 1) \approx 3.4402 \times 10^{-6}.$$

Thus the adjacent middle/high pair is worse at this fixed mean.

Numerical roots of the pairwise loss differences in the baseline $\rho = 1, \eta = 0.8$ model show that the identity of the maximizing pair changes near

$$m \approx 0.553191 \quad \text{and} \quad m \approx 6.729134.$$

The low/high pair dominates over most of the interior and still produces the global baseline maximum, but there is no general pointwise low/high theorem. The reason is structural: J depends on the predictive distribution $r = \pi P$ and the signal experiment, not on the dispersion of the current impact support alone.

9 Discussion

9.1 Forecast Error versus Adaptation Value

The affine condition (2) is useful because it removes a simpler source of policy disagreement. If two equal-mean beliefs generated different expected future impact paths, different actions would not be surprising. Corollary 1 eliminates that channel. In the matched environment, different actions arise because the same expected future liquidity can carry different option value for adapting to future information.

9.2 What the CE Loss Measures

Equation (10) is specific to the current CE approximation. It does not measure the loss of a controller that permanently replaces the belief state by a scalar and then attempts to filter using that scalar alone. In general, the next posterior mean after a signal is not recoverable from the previous posterior mean alone. The interpretation is instead modular: a filtering layer may maintain the full posterior, while the execution decision interface is restricted to a point estimate. The result quantifies the current decision loss from that interface compression.

9.3 Magnitude

The CE value losses in the baseline example are small in normalized units. That is not inconsistent with visible policy differences. Since $\Delta_x = O(J)$ while $R_{\text{CE}} = O(J^2)$, the objective is locally flat around the optimum. The paper's claim is therefore structural rather than that the baseline parameterization produces economically large dollar losses.

9.4 Scope

The model is intentionally minimal. It omits permanent impact, endogenous signal precision, action-dependent learning, and continuous-time filtering. Adding these features would make the control problem richer but is not needed for the fixed-estimate question isolated here.

A four-state extension is likewise unnecessary for the central result. With three distinct fixed impact states, normalization, mean, and second moment identify the posterior through a Vandermonde system. Four or more states would be needed only to ask the different question of whether mean and variance together are sufficient.

9.5 Companion Visualization and Browser Implementation

Figure 1 is the paper’s archival rendering of the baseline geometry. It is deterministic: the simplex, the CE-loss field, the five topology-guided contour levels, and the fixed-mean fiber are direct outputs of the model. The selected levels represent the five stable connected-component regimes of the baseline loss surface and therefore produce nine visible contour components in total. The companion browser piece¹ uses the same contour levels and geometry but presents them as a generative construction. The triangle, internal lattice, and contour components are drawn in time, after which a seeded Bayesian belief-path realization is overlaid. During that realization, the current posterior is labeled π_t and the current fixed-mean fiber is labeled by $m(\pi_t)$, while π^* remains the fixed global-loss landmark. The benchmark beliefs π^C and π^U remain part of the paper’s static figure and numerical example but are not displayed in the browser realization.

That browser implementation is expository rather than evidentiary. The theorem and numerical claims in the paper do not depend on the website animation, and the website animation does not introduce additional empirical content beyond the model already analyzed here. Its role is to make the central distinction visually legible: a common scalar liquidity estimate can sit atop a richer posterior geometry with decision-relevant structure that the scalar alone does not preserve.

10 Conclusion

A posterior mean can preserve an expected liquidity forecast without preserving the execution decision. In the latent-liquidity model studied here, affine mean dynamics allow two beliefs to share not only the same current expected impact but the same unconditional expected impact at every future horizon. Informative signals can still separate their optimal trades because posterior shape changes the value of future adaptation.

At the first learning-sensitive horizon, this effect is summarized by the information premium $J(\pi)$. The corresponding CE action distortion and value loss are available in closed form. Under a symmetric finite-state signal experiment, the information premium is convex along fixed-mean belief sets, which reduces the worst-case CE loss for a reported liquidity estimate at this horizon to a finite comparison of two-state posterior mixtures.

The result is not a new general compression theorem. Its contribution is the execution-specific quantification: it identifies how a scalar liquidity estimate can conceal decision-relevant information and gives an exact, computable worst-case loss for the CE execution decision induced by that compression at the first learning-sensitive horizon.

A Small-Information Expansion

For $\eta < 1$, define

$$t = \frac{d}{c} = \frac{n\eta - 1}{1 - \eta}.$$

Then

$$c = \frac{1}{n + t}, \quad d = \frac{t}{n + t},$$

and $t = 0$ is the uninformative experiment. In the reciprocal term of (15),

$$\frac{(c + dr_y)^2}{cD_\mu + dr_yD_y} = \frac{(1 + tr_y)^2}{(n + t)(D_\mu + tr_yD_y)}.$$

¹Available at <https://arian.ms/work.html#same-estimate-work> under “Same Estimate, Different Information.”

Using $\sum_y r_y = 1$ and $\sum_y r_y D_y = D_\mu$, the constant term sums to $1/D_\mu$ and the first-order term cancels. Moreover,

$$\frac{1}{n+t} \sum_y \frac{(1+tr_y)^2}{D_\mu + tr_y D_y} = \frac{1}{D_\mu} + \frac{s^2 t^2}{n D_\mu^3} \sum_y r_y^2 (\mu - \lambda_y)^2 + O(t^3), \quad (21)$$

where $D_\mu - D_y = s(\mu - \lambda_y)$. Substitution into (15) gives

$$J(\pi) = \frac{A^2 t^2}{n D_\mu^3} \sum_y r_y^2 (\mu - \lambda_y)^2 + O(t^3). \quad (22)$$

Hence

$$J = O(t^2), \quad \Delta_x = O(t^2), \quad R_{\text{CE}} = O(t^4).$$

The expansion is useful for interpreting the weak-information regime.

B A Constructive Affine-Transition Family

For a three-state model with

$$\lambda_1 < \lambda_2 < \lambda_3,$$

for parameters $\beta \in [0, 1]$ and $s_0 \in [0, 1]$, one convenient row-stochastic family satisfying the affine condition is

$$P = \begin{pmatrix} \beta & 1-\beta & 0 \\ b & 1-s_0 & c_0 \\ 0 & 1-\beta & \beta \end{pmatrix},$$

where

$$b = s_0 \frac{\lambda_3 - \lambda_2}{\lambda_3 - \lambda_1}, \quad c_0 = s_0 \frac{\lambda_2 - \lambda_1}{\lambda_3 - \lambda_1}.$$

Then

$$P\lambda = (1-\beta)\lambda_2 \mathbf{1} + \beta\lambda.$$

The baseline matrix corresponds to $\beta = 0.7$ and $s_0 = 0.25$ for $\lambda = (0.5, 2, 8)$.

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